👩‍🏫 Staff Guide · Chemistry · Unit 0 Lab Safety & Equipment (Level 3, tester)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow Mr. Brittle's slides with Level 3 guided notes and readings.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Syllabus & Safety Contract + Safety Presentation (🚧 not in the export)	Level 3 reading “Working Safely in the Chemistry Lab” + notes part A (incl. 🚧 Find it in our room)
L0.1 Lab Equipment Activity (ClEvR poster)	Notes part C: equipment table + one ClEvR statement (same as the class)
L0.2 Elements Reacting (demos and lab)	Do the class lab with goggles; notes part B hazard symbols supports the HCl warning
L0.3 Equipment Practice (8 stations)	Notes parts B-C: hazard table, units table, water displacement math; quick check + Level 3 game

2. Supports built in

- Reading and notes have an online copy with a 🔊 Read aloud button
- Key-word box at the top of the notes; worked example before the math
- No slide decks: follow Mr. Brittle's Safety Presentation and station slides

3. What to grade

- Guided notes stand in for the L0.1 and L0.3 sheets (same points when complete)
- Quick check: score, then scale to the class points
- Safety contract: 🚧 ask Mr. Brittle how SLE students sign it
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with Mr. Brittle”.

🚧 Under construction (check with Mr. Brittle)

- **Where the safety equipment is in his room**: We do not know where the goggles, eyewash, safety shower, fire extinguisher, fire blanket, first aid kit and exits are in Mr. Brittle's room. Staff: walk the room and fill in the “Find it in our room” box.
- **Safety Presentation**: His Week 2 “Safety Presentation” is not in the export. Our slides are built from his Safety Contract rules.
- **Safety Contract**: Our notes do not replace his signed Syllabus & Safety Contract. Ask if SLE students sign his contract (read aloud) or a shorter one.
- **L0.3 stations**: Ask which of the 8 L0.3 stations Level 1-3 students should do. Stations 7-8 use a hot plate and NaOH (corrosive); staff should be at the station.

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🥽 Lab Safety & Equipment: Guided Notes

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Use the class slides and your reading. Key words:

goggles | *eyewash station* | *safety shower* | *hazard symbol* | *corrosive* | *flammable* | *graduated cylinder* | *meniscus* | *balance* | *tare* | *Celsius*

A. Safety rules (Safety Contract)

1. Wear goggles any time there are chemicals, heat or _____.
2. If a chemical gets in your eyes, rinse for at least _____ minutes.
3. If someone is hurt, yell "_____, Code one."
4. Find it in our room 🚧: write where each one is in Mr. Brittle's room: goggles, eyewash, safety shower, fire extinguisher, exits.

5. Pick one rule from the contract. Explain how it keeps people safe.

B. Hazard symbols (L0.3 Station 6)

6. Fill in the table.

Hazard word	What it means	One way to stay safe
Flammable		
Corrosive		
Toxic		
Irritant		

C. Equipment and measuring (L0.1, L0.3)

7. Fill in the table.

Equipment	What it measures or does	Unit
Graduated cylinder		
Electronic balance		
Thermometer		

8. **Volume by water displacement (L0.3 Station 1)**

> Start: 100 mL of water. Drop in a marble. New level: 106 mL.
> Volume of marble = 106 − 100 = 6 mL.

a) Start 50 mL, new level 58 mL. Volume of the bead = ____ mL _____

b) Start 72 mL, new level 75 mL. Volume = ____ mL _____

9. ClEvR (L0.1): Write a claim, evidence and reasoning for one tool. “The ___ is used for ___ because it ___, and that means ___.”

🥽 Lab Safety & Equipment: Reading

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Working Safely in the Chemistry Lab

Chemistry is a hands-on class, and many labs use chemicals that can be dangerous. Mr. Brittle's Safety Contract lists the rules. You must follow them at all times. The most important rule is simple: follow all written and spoken directions, and ask before you do anything you are not sure about.

Protect your body. Wear goggles any time chemicals, heat or glassware are used. There are no exceptions. Tie back long hair, secure loose clothing and jewelry, wear shoes that cover your whole foot, and wear a lab apron. Never eat, drink or chew gum in the lab, and never use lab glassware as a cup.

Know what to do in an accident. Report every spill, break, cut or burn, no matter how small. If you or your partner is hurt, yell “Code one, Code one.” If a chemical splashes in your eyes or on your skin, rinse it with running water at the eyewash station or safety shower for at least 20 minutes. Never pick up broken glass with your hands; use a brush and dustpan.

Handle chemicals with care. Treat every chemical as dangerous. Read the label twice, take only what you need, and never put unused chemicals back in the bottle. Hazard symbols on the label tell you the danger: flammable (can catch fire), corrosive (destroys skin or metal), toxic (poison), irritant (makes skin red or itchy), and oxidizing (releases oxygen, which feeds a fire).

Use the right tool to measure. A graduated cylinder measures the volume of a liquid in milliliters (mL); read the bottom of the curved surface, called the meniscus. An electronic balance measures mass in grams (g); press tare to set it to zero before you add your sample. A thermometer measures temperature, and scientists use degrees Celsius (°C). A beaker holds and mixes liquids, but its marks are only an estimate.

Check your reading

1. Why must you wear goggles even when you are only using glassware?

2. Why should you never return unused chemicals to their bottle?

3. Why is a graduated cylinder better than a beaker for measuring 25 mL of water?

🥽 Lab Safety & Equipment: Quick Check

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When must you wear goggles?

- A. Only when using fire
- B. Only during tests
- C. Any time chemicals, heat or glassware are used

2. A chemical splashes on your skin. What do you do first?

- A. Rinse with running water for 20 minutes
- B. Wipe it with your shirt
- C. Wait to see if it hurts

3. A label shows the corrosive symbol. The chemical can…

- A. Glow in the dark
- B. Only catch fire
- C. Destroy skin or metal

4. Where do you read the volume in a graduated cylinder?

- A. The bottom of the meniscus
- B. The top edge of the curve
- C. The side of the cylinder

5. Why do you press tare on a balance with a paper boat on it?

- A. To change grams to pounds
- B. So only the sample's mass is shown
- C. To turn off the balance

6. Start: 40 mL of water. After a rock is added: 47 mL. What is the rock's volume? Show your work.

7. Your lab partner drops a beaker and it breaks. List 2 things you should do.

🥽 Lab Safety & Equipment: Guided Notes · ANSWER KEY

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Use the class slides and your reading. Key words:

goggles | *eyewash station* | *safety shower* | *hazard symbol* | *corrosive* | *flammable* | *graduated cylinder* | *meniscus* | *balance* | *tare* | *Celsius*

A. Safety rules (Safety Contract)

1. Wear goggles any time there are chemicals, heat or **glassware**.
2. If a chemical gets in your eyes, rinse for at least **20** minutes.
3. If someone is hurt, yell "**Code one**, Code one."
4. Find it in our room 🚧: write where each one is in Mr. Brittle's room: goggles, eyewash, safety shower, fire extinguisher, exits.
Answers depend on the room (staff: check with Mr. Brittle).
5. Pick one rule from the contract. Explain how it keeps people safe.
Example: tie back long hair, so it cannot catch fire or fall into chemicals.

B. Hazard symbols (L0.3 Station 6)

6. Fill in the table.

Hazard word	What it means	One way to stay safe
Flammable	**can catch fire**	**keep away from flames and heat**
Corrosive	**destroys skin or metal**	**goggles, gloves, apron**
Toxic	**poison**	**never taste; wash hands**
Irritant	**makes skin red or itchy**	**do not touch; gloves**

C. Equipment and measuring (L0.1, L0.3)

7. Fill in the table.

Equipment	What it measures or does	Unit
Graduated cylinder	**volume of a liquid**	**mL**
Electronic balance	**mass**	**g**
Thermometer	**temperature**	**°C**

8. **Volume by water displacement (L0.3 Station 1)**

> Start: 100 mL of water. Drop in a marble. New level: 106 mL.
> Volume of marble = 106 − 100 = 6 mL.

a) Start 50 mL, new level 58 mL. Volume of the bead = ____ mL **8**

b) Start 72 mL, new level 75 mL. Volume = ____ mL **3**

9. ClEvR (L0.1): Write a claim, evidence and reasoning for one tool. “The ___ is used for ___ because it ___, and that means ___.”

Example: A test tube is used for holding or heating small amounts because it is a small, narrow Pyrex tube with no marks, so it is not for measuring but can be heated.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*

🥽 Lab Safety & Equipment: Quick Check · ANSWER KEY

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When must you wear goggles?

A. Only when using fire	B. Only during tests	**C. Any time chemicals, heat or glassware are used**

2. A chemical splashes on your skin. What do you do first?

A. Rinse with running water for 20 minutes	B. Wipe it with your shirt	C. Wait to see if it hurts

3. A label shows the corrosive symbol. The chemical can…

A. Glow in the dark	B. Only catch fire	**C. Destroy skin or metal**

4. Where do you read the volume in a graduated cylinder?

A. The bottom of the meniscus	B. The top edge of the curve	C. The side of the cylinder

5. Why do you press tare on a balance with a paper boat on it?

A. To change grams to pounds	**B. So only the sample's mass is shown**	C. To turn off the balance

6. Start: 40 mL of water. After a rock is added: 47 mL. What is the rock's volume? Show your work.
47 − 40 = 7 mL.

7. Your lab partner drops a beaker and it breaks. List 2 things you should do.
Tell the teacher right away; do not touch the glass; use a brush and dustpan; put it in the glass disposal box.

🔑 *Written by our team from his Safety Contract and L0.3 Equipment Practice, please check.*